

Kenta Suzuki

✉ kjsuzuki@princeton.edu <https://kentajsuzuki.github.io/>

Research Interests

Algebraic Geometry, Representation Theory

Education

Princeton University

Doctorate in Mathematics

Princeton, NJ

Anticipated Graduation: May 2030

Massachusetts Institute of Technology

Bachelors of Science in Mathematics

Cambridge, MA

Graduated May 2025

- **GPA:** 5.0/5.0
- **Coursework:** Categorical Langlands and Shimura Varieties, Topological modular forms, Motives at p , Infinite-dimensional Lie algebras, Cohomology of p -adic analytic groups, Symplectic geometry in characteristic p .

Preprints

- K. Suzuki, *Geometric characterization of the group law in the Weyl group*, arXiv:2407.02752 *Mar 25, 2025*
- K. Suzuki, *Trace Paley-Wiener theorem for Braverman-Kazhdan's asymptotic Hecke algebra*, arXiv:2407.02752 *Jul 3, 2024*
- V. Krylov and K. Suzuki, *Subregular nilpotent orbits and explicit character formulas for modules over non-simply laced affine Lie algebras*, to appear in Mathematical Research Letters *Jan 12, 2024*
- K. Suzuki, *Rationality of the Local Jacquet-Langlands Correspondence for $GL(n)$* , arXiv:2307.06039 *Jul 12, 2023*
- K. Suzuki and Y. Xu, *The explicit Local Langlands Correspondence for G_2 II: character formulas and stability*, arXiv:2304.02630, submitted. *Apr 2023*
- K. Suzuki and Y. Xu, *The explicit Local Langlands Correspondence for GSp_4, Sp_4 , and stability (with an application to Modularity Lifting)*, arXiv:2304.02622, submitted. *Apr 2023*
- K. Suzuki, *Gelfand-Kirillov dimension of representations of GL_n over a non-archimedean local field*, arXiv:2208.05139, submitted. *Aug 2022*
- K. Suzuki and M. Zieve, *Meromorphic functions with the same preimages at several finite sets*, PRIMES website. *Aug 2022*

Research Talks

- NUS:** Fargues's categorical conjecture for elliptic parameters for $SL(n)$ *Jun, 2025*
- CASL:** Fargues's categorical conjecture for elliptic parameters for $SL(n)$ *Jun, 2025*
- Stanford:** Fargues's categorical conjecture for elliptic parameters for $SL(n)$ *May 1, 2025*
- Academia Sinica:** Cocenter of Braverman-Kazhdan's Asymptotic Hecke Algebra *Dec 16, 2024*
- Yale:** Trace Paley-Wiener theorem for Braverman-Kazhdan's Asymptotic Hecke Algebra *Sep 16, 2024*
- MIT:** Affine Kazhdan-Lusztig polynomials on the subregular cell *Feb 23, 2024*
- MIT:** The explicit Local Langlands Correspondence for G_2 and GSp_4 *Oct 4, 2023*
- SPUR:** Gelfand-Kirillov dimension of representations of p -adic GL_n *Aug 5, 2022*

Expository Talks

STAGE: Weights and the statement of Weil II	<i>Nov 20, 2025</i>
STAGE: Monodromy of the Kodaira–Parshin family	<i>May 8, 2025</i>
de Rham cohomology via stacks	<i>Oct 22 and Nov 1, 2024</i>
STAGE: Relationship of integrable connections to PDEs and nth order ODEs	<i>Sep 30, 2024</i>
STAGE: Criterion for the non-existence of rational p-torsion	<i>Nov 16, 2023</i>
Modularity/Fermat Seminar: Taylor-Wiles Deformations	<i>Nov 3, 2023</i>
K3 surface seminar: Orthogonal Shimura varieties and moduli of K3 surfaces	<i>Oct 16, 2023</i>
Modularity/Fermat Seminar: Langlands-Tunnell, Part I	<i>Sep 22, 2023</i>
STAGE: Complex tori and abelian varieties	<i>Feb 13, 2023</i>
STAGE: Review of Étale Cohomology	<i>Sep 20, 2022</i>

Teaching and Mentoring

MIT PRIMES (Program for Research in Mathematics, Engineering, and Science)

<i>Research mentor</i>	<i>Nov 2023 -</i>
• Mentoring high-school student on computing affine Kazhdan-Lusztig polynomials and on the Temperley-Lieb algebra. • Student my first year was semi-finalist in the S.-T. Yau Competition, my second year won Silver in the S.-T. Yau Competition.	

SPUR+ Mentor

<i>Research mentor</i>	<i>June 2026 - August 2026</i>
• Mentored two undergraduate students on a project on the Gelfand-Tsetlin algebra in positive characteristic.	

DRP (Directed Reading Program)

<i>Research mentor</i>	<i>Sep 2025 -</i>
• Mentored 4 undergraduate students on the knot theory, combinatorics, linear algebra, and algebraic geometry.	

RSI (Research Science Institute)

<i>Research mentor</i>	<i>Jun 2025 - Aug 2025</i>
• Mentored high-school students on knot theory and the Temperley-Lieb algebra. • Students won Top 300 in Regeneron's Science Talent Search.	

Awards and Honors

George Lusztig Mentor for PRIMES (MIT)	<i>Dec 2024</i>
Frank and Brennie Morgan Prize for Outstanding Research in Mathematics (national)	<i>Oct 2024</i>
Goldwater Scholar (national)	<i>Mar 2024</i>
Mitsui Mirai Research Grant	<i>2024 - present</i>
ANRI Fundamental Science Research Fellowship	<i>Nov 2023</i>
• Research and travel grant.	
Hartley Rogers Jr. Prize for SPUR (MIT)	<i>Aug 2022</i>
William Lowell Putnam Competition Honorable Mention (national)	<i>Dec 2021 and Dec 2023</i>
Davidson Fellow Scholarship Honorable Mention (national)	<i>2021</i>
Ezoe Memorial Recruit Foundation Scholar	<i>2021 - present</i>

- Full tuition merit scholarship, stipend, and travel grant.

Masason Foundation Scholar

2018 - present

- Full tuition merit scholarship, research and travel grant.

Seminars

Seminar on Bezrukavnikov's equivalence

Co-Organizer

Spring 2026

- I co-organized a seminar, with Bhargav Bhatt, Jacob Lurie, and Mingjia Zhang, on Bezrukavnikov's equivalence.

Quantum Topology Seminar

Organizer

Fall 2025

- I organized a seminar on quantum topology at Princeton in the Fall of 2025.

First-year Seminar

Co-Organizer

Fall 2025

- I co-organized the seminar for first-year graduate students at Princeton.

STAGE (Seminar on Topics in Arithmetic, Geometry, Etc.)

Co-Organizer

Spring 2023, Fall 2024

- The STAGE seminar is a learning seminar for graduate students in number theory. The topic for Spring 2023 was *Shimura varieties*, and the topic for Fall 2024 is *Algebraic differential equations*.
- Planned the talk topics, collected speakers, and organized post-seminar social events.